



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

INSPIRING CREATIVE AND INNOVATIVE MINDS

www.utm.my

Chapter 2 (Part 1)

Relations



Relations

- A (binary) relation R from a set X to a set Y is a subset of the Cartesian product $X \times Y$.
- If $(x, y) \in R$, we write

$$x R y \quad (x \text{ is related to } y)$$

(Binary) relation from X to Y , where $x \in X$, $y \in Y$,
 $(x, y) \in X \times Y$ and $R \subseteq X \times Y$

$$x R y \Leftrightarrow (x, y) \in R$$



Example

- $A = \{ 1, 2, 3, 4 \} , B = \{ p, q, r \}$
- $R = \{ (1, q), (2, r), (3, q), (4, p) \}$
- $R \subseteq A \times B$
- R is the relation from A to B
 $1 R q$ $3 \not R p$

$A \times B$

$(1, p), (1, q), (1, r),$
 $(2, p), (2, q), (2, r),$
 $(3, p), (3, q), (3, r),$
 $(4, p), (4, q), (4, r)$



Example

$$aRb \leftrightarrow a-b \in \mathbb{Z}^{\text{even}}$$

- *Finite set:* $A = \{ 1, 2 \}$, $B = \{ 1, 2, 3 \}$

$$R = \{ (1, 1), (2, 2), (1, 3) \}$$

- *Infinite set:* $A = \mathbb{Z}$ and $B = \mathbb{Z}$

$$R = \{ \dots(-3, -1), (-2, 2), (1, 3), \dots \}$$

(note: $\mathbb{Z}$ is set of integers)

Example

- $A = \{ \text{New Delhi, Ottawa, London, Paris, Washington} \}$
- $B = \{ \text{Canada, England, India, France, United States} \}$
- Let $x \in A$, $y \in B$. Define the relation between x and y by
“ x is the capital of y ”
- $R = \{ (\text{New Delhi, India}), (\text{Ottawa, Canada}),$
 $(\text{London, England}), (\text{Paris, France}),$
 $(\text{Washington, United States}) \}$



Example

- “less than” relation from $A=\{0, 1, 2\}$ to $B=\{1, 2, 3\}$

- Traditional notation:

$0 < 1, 0 < 2, 0 < 3, 1 < 2, 1 < 3, 2 < 3$

- Set notation

$$A \times B = \{ (0,1), (0,2), (0,3), (1,1), (1,2), (1,3), (2,1), (2,2), (2,3) \}$$

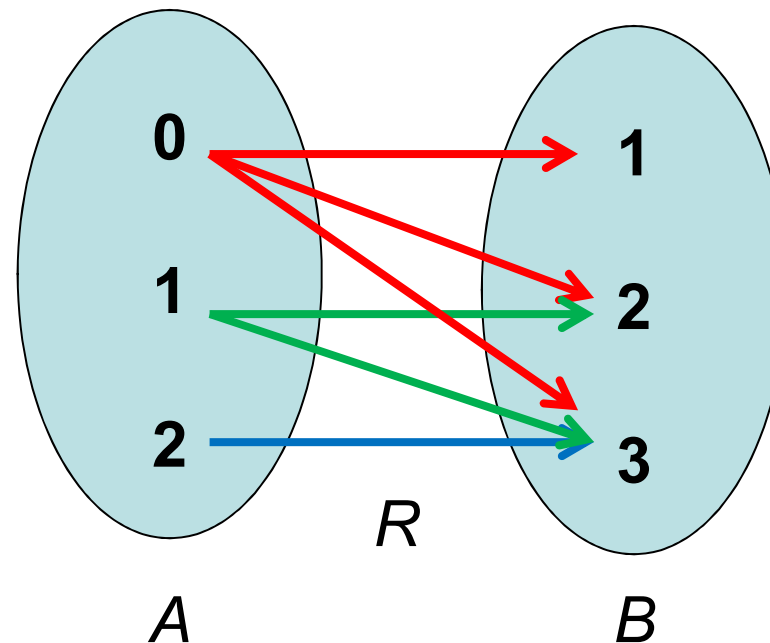
$$R = \{ (0,1), (0,2), (0,3), (1,2), (1,3), (2,3) \}$$



example

$$R = \{ (0,1), (0,2), (0,3), (1,2), (1,3), (2,3) \}$$

■ Arrow diagrams





Domain and Range

- Let R , a relation from A to B .
- The set, $\{ a \in A \mid (a, b) \in R \text{ for some } b \in B \}$ is called the **domain** of R .
- The set, $\{ b \in B \mid (a, b) \in R \text{ for some } a \in A \}$ is called the **range** of R .
- In case $A=B$, we call R a(binary) **relation on** A .



Example

- Let R be a relation on $X = \{1, 2, 3, 4\}$ defined by $(x,y) \in R$ if $x \leq y$, and $x, y \in X$.
- Then,
 $R = \{ (1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4) \}$
- The domain and range of R are both equal to X .



Example

- Let $X = \{ 2, 3, 4 \}$ and $Y = \{ 3, 4, 5, 6, 7 \}$
If we define a relation R from X to Y by,
 $(x,y) \in R$ if x divides y (with zero remainder)

- We obtain,

$$R = \{ (2,4), (2,6), (3,3), (3,6), (4,4) \} \quad (x, y)$$

The domain of R is $\{2,3,4\}$

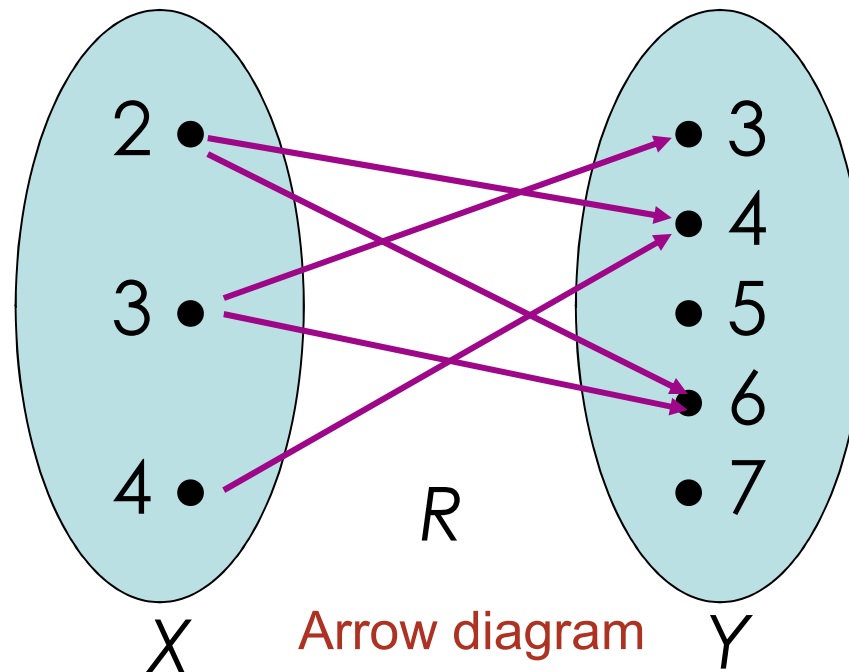
The range of R is $\{3,4,6\}$

x - domain
y - range



example

$$R = \{ (2,4), (2,6), (3,3), (3,6), (4,4) \}$$





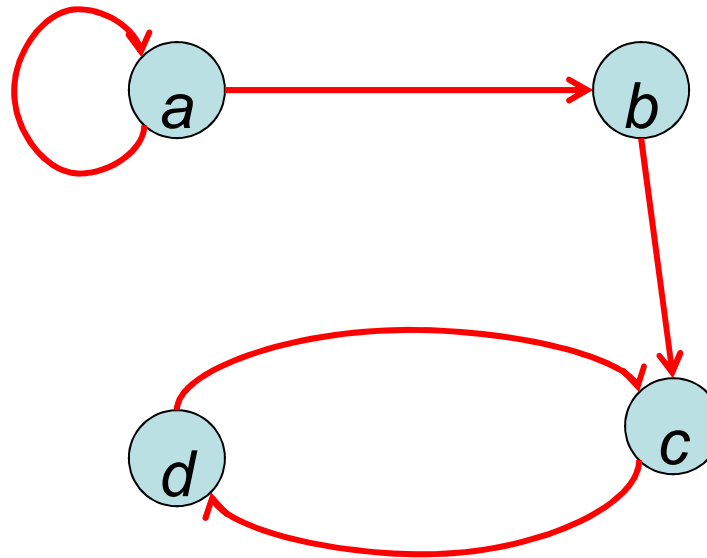
Digraph

- An informative way to picture a relation on a set is to draw its digraph.
- Let R be a relation on a finite set A .
- Draw dots (**vertices**) to represent the elements of A .
- If the element $(a,b) \in R$, draw an arrow (called a **directed edge**) from a to b .



example

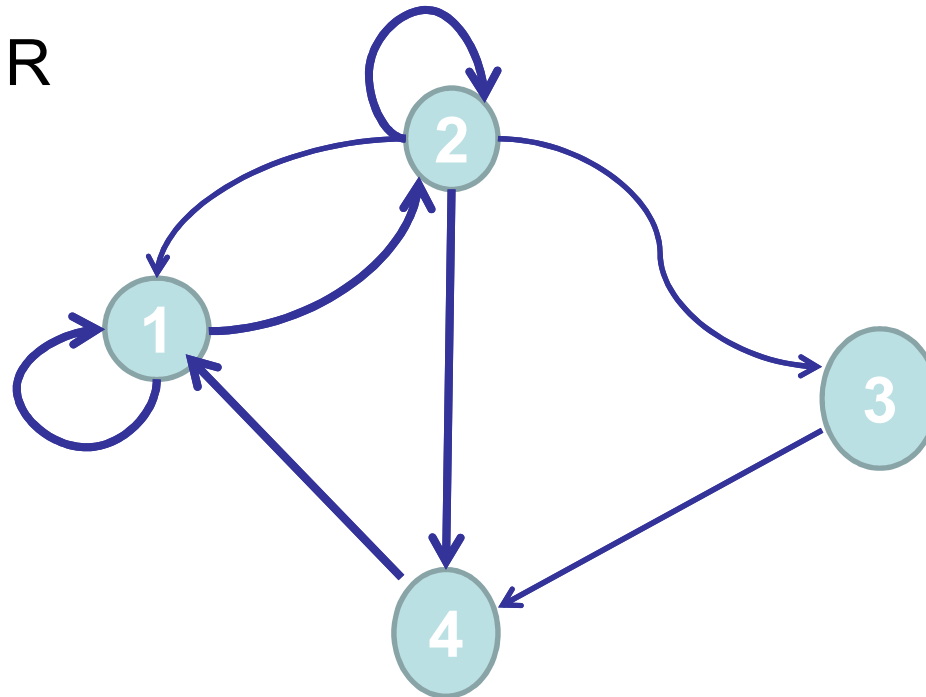
- The relation R on $A = \{a, b, c, d\}$,
 $R = \{(a, a), (a, b), (c, d), (d, c), (b, c)\}$





example

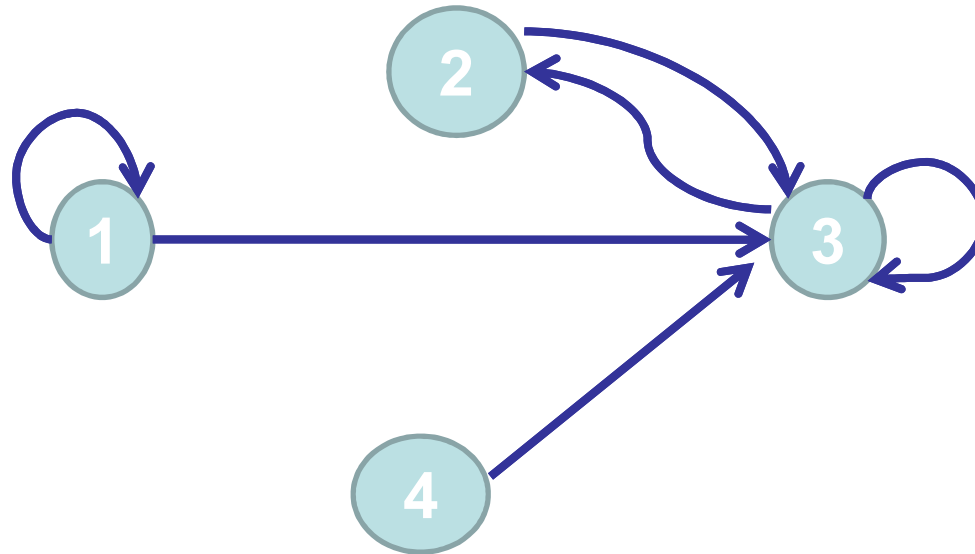
- Let, $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4), (4, 1)\}$
- Draw the digraph of R





Example

Find the relation determined by digraph below.



- Since $a R b$ if and only if there is an edge from a to b , so
 $R = \{ (1,1), (1,3), (2,3), (3,2), (3,3), (4,3) \}$

Matrices of Relations

- A matrix is a convenient way to represent a relation R from A to B .
- Label the rows with the elements of A (in some arbitrary order)
- Label the columns with the elements of B (in some arbitrary order)



Matrices of Relations

- Let $A = \{a_1, a_2, \dots, a_n\}$ and $B = \{b_1, b_2, \dots, b_p\}$ be finite nonempty sets.
- Let R be a relation from A into B .
- Let $M_R = [m_{ij}]_{n \times p}$ be the Boolean $n \times p$ matrix, where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{otherwise} \end{cases}$$



Matrices of Relations

$$M_R = \begin{bmatrix} m_{11} & m_{12} & .. & .. & m_{1p} \\ m_{21} & m_{22} & .. & .. & m_{2p} \\ : & : & .. & .. & : \\ : & : & .. & .. & : \\ m_{n1} & m_{n2} & .. & .. & m_{np} \end{bmatrix}$$



- Let $A = \{1, 3, 5\}$ and $B = \{1, 2\}$
- Let R be a relation from A to B and $R = \{(1,1), (3,2), (5,1)\}$
- Then the matrix represent R is

$$\begin{array}{c} \\ 1 \\ 3 \\ 5 \end{array} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$



example

- The relation,

$$R = \{ (1,b), (1,d), (2,c), (3,c), (3,b), (4,a) \}$$

from, $X = \{ 1, 2, 3, 4 \}$ to $Y = \{ a, b, c, d \}$

	a	b	c	d
1	0	1	0	1
2	0	0	1	0
3	0	1	1	0
4	1	0	0	0

or

	d	b	a	c
2	0	0	0	1
3	0	1	0	1
4	0	0	1	0
1	1	1	0	0



example

- The matrix of the relation R from $\{2, 3, 4\}$ to $\{5, 6, 7, 8\}$ defined by

$x R y$ if x divides y

	5	6	7	8
2	0	1	0	1
3	0	1	0	0
4	0	0	0	1



example

- Let $A = \{ a, b, c, d \}$
- Let R be a relation on A .
- $R = \{ (a,a), (b,b), (c,c), (d,d), (b,c), (c,b) \}$

	a	b	c	d
a	1	0	0	0
b	0	1	1	0
c	0	1	1	0
d	0	0	0	1



In Degree and Out Degree

- If R is a relation on a set A and $a \in A$, then the **in-degree** of a (relative to relation R) is the number of $b \in A$ such that $(b, a) \in R$.
- The **out degree** of a is the number of $b \in A$ such that $(a, b) \in R$



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

In Degree and Out Degree

- Meaning that, in terms of the digraph of R , is that the in-degree of a vertex is
“the number of edges terminating at the vertex”
- The out-degree of a vertex is
“the number of edges leaving the vertex”



Example

- Let $A = \{a, b, c, d\}$, and let R be the relation on A that has the matrix (given below)

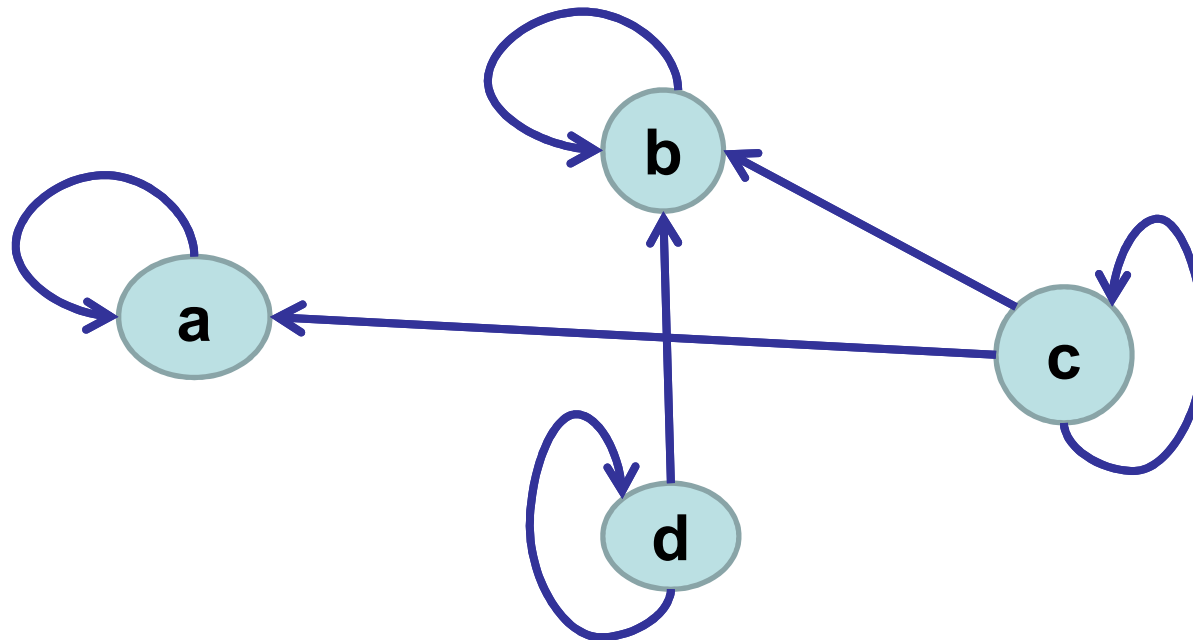
$$M_R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

- Construct the digraph of R , and list in-degrees and out-degrees of all vertices.



Example

	a	b	c	d
In-degree	2	3	1	1
Out-degree	1	1	3	2





Reflexive Relations

- A relation R on a set X is called **reflexive** if $(x,x) \in R$ for every $x \in X$.
 - That is, if xRx for all $x \in X$.
- (R is reflexive if every element $x \in X$ is related to itself)

Example

- The relation R on $X = \{1, 2, 3, 4\}$ defined by $(x, y) \in R$ if $x \leq y$, $x, y \in X$ is reflexive because for each element $x \in X$, $(x, x) \in R$
- $(1, 1), (2, 2), (3, 3), (4, 4)$ are each in R .



example

- The relation,

$$R = \{ (a,a), (b,c), (c,b), (d,d) \}$$

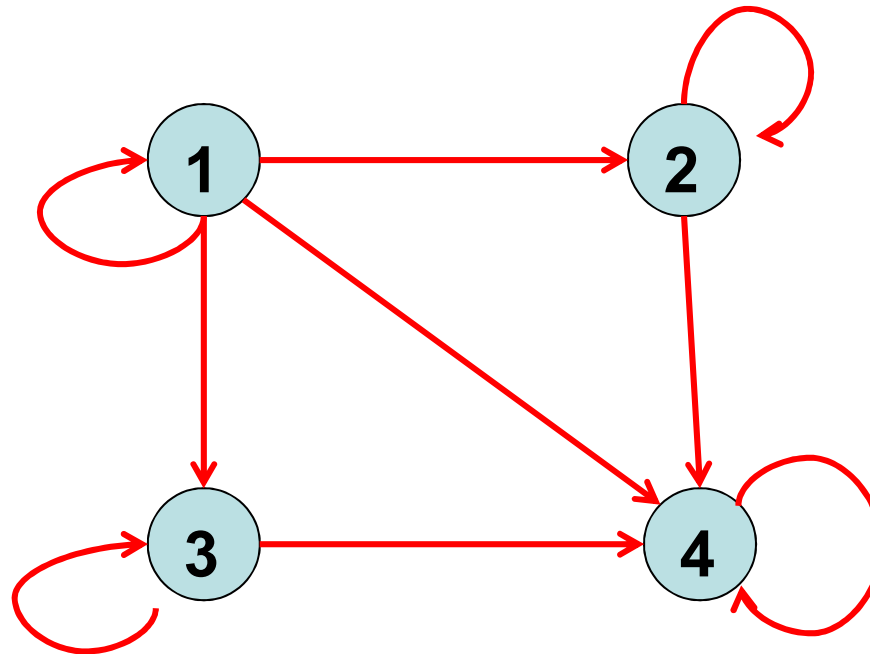
on $X = \{a, b, c, d\}$ is not reflexive.

- For example, $b \in X$, but $(b,b) \notin R$



Reflexive Relations

- The digraph of a reflexive relation has a loop at every vertex.
- example





UTM
UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Reflexive Relations

■ Irreflexive

- A relation R on a set A is **irreflexive** if $x \not R x$ or $(x, x) \notin R; \forall x: x \in X$

■ Not Reflexive

- A relation R is **not reflexive** if at least one pair of $(x, x) \notin R, \forall x: x \in X$

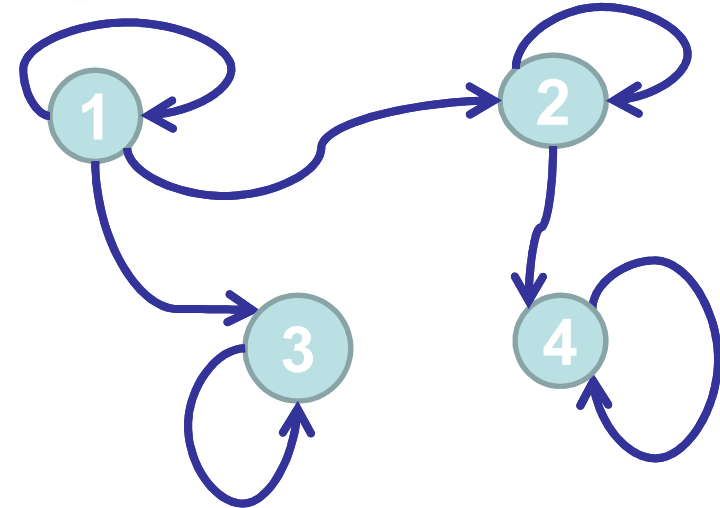


UTM
UNIVERSITI TEKNOLOGI MALAYSIA

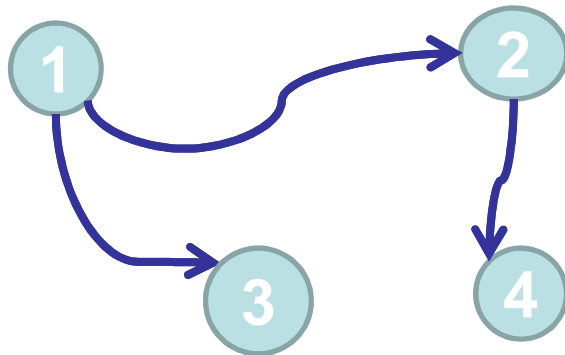
www.utm.my

example

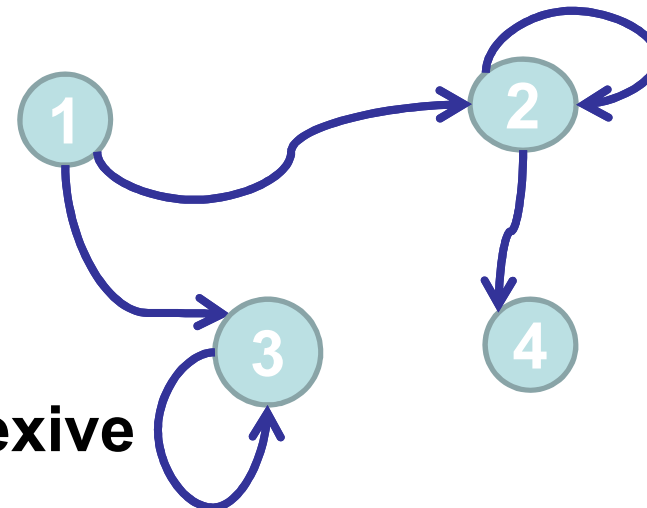
Reflexive



Irreflexive



Not reflexive





UTM
UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

- Consider the following relations on the set $\{1, 2, 3\}$

$$R_1 = \{ (1,1), (1,2), (2,1), (2,2), (3,1), (3,3) \}$$

$$R_2 = \{ (1,1), (1,3), (2,2), (3,1) \}$$

$$R_3 = \{ (2,3) \}$$

$$R_4 = \{ (1,1) \}$$

Which of them are reflexive?



Reflexive Relations

- The relation R is reflexive if and only if the matrix of relation has 1's on the main diagonal.

- example

	a	b	c	d
a	1	0	0	0
b	0	1	1	0
c	0	1	1	0
d	0	0	0	1



Reflexive Relations

The relation R is *irreflexive* if and only if the matrix relation have all 0's on its main diagonal

■ example

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$



example

- The relation R is **not reflexive**.

	a	b	c	d
a	1	0	0	0
b	0	0	1	0
c	0	1	1	0
d	0	0	0	1

$b \in X$
 $(b, b) \notin R$



Symmetric Relations

- A relation R on a set X is called symmetric if for all $x, y \in X$, if $(x, y) \in R$, then $(y, x) \in R$.

$$\forall x, y \in X, (x, y) \in R \rightarrow (y, x) \in R$$

- Let M be the matrix of relation R .
The relation R is symmetric if and only if for all i and j , the ij th entry of M is equal to the ji th entry of M .



Symmetric Relations

- The matrix of relation M_R is symmetric if $M_R = M_R^T$

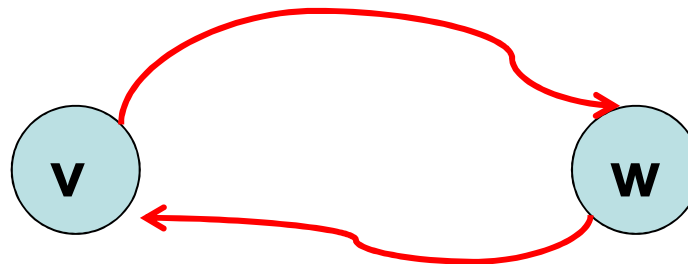
- Example

$$M_R = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix} = M_R^T$$



Symmetric Relations

- The digraph of a symmetric relation has the property that whenever there is a directed edge from v to w , there is also a directed edge from w to v .





example

- The relation $R = \{ (a,a), (b,c), (c,b), (d,d) \}$ on $X = \{ a, b, c, d \}$

$(b,c) \in R$
 $(c,b) \in R$

	a	b	c	d
a	1	0	0	0
b	0	0	1	0
c	0	1	0	0
d	0	0	0	1

symmetric



example

- The relation R on $X = \{ 1, 2, 3, 4 \}$, defined by
 $(x,y) \in R$ if $x \leq y, x,y \in X$

$(2,3) \in R$
 $(3,2) \notin R$

	1	2	3	4
1	1	1	1	1
2	0	1	1	1
3	0	0	1	1
4	0	0	0	1

not symmetric



Antisymmetric Relations

- A relation, R on a set X is called antisymmetric, if for all $x, y \in X$, if $(x, y) \in R$ and $x \neq y$, then $(y, x) \notin R$.
- A relation R on set X is antisymmetric if $x \neq y$, whenever xRy , then $y \not R x$. In other word if whenever xRy , then yRx then it implies that $x=y$

$$\forall x, y \in A, (x, y) \in R \wedge x \neq y \rightarrow (y, x) \notin R$$

Or

$$\forall x, y \in A, (x, y) \in R \wedge (y, x) \in R \rightarrow x = y$$



Antisymmetric Relations

- Matrix $M_R = [M_{ij}]$ of an antisymmetric relation R satisfies the property that if $i \neq j$, then $m_{ij}=0$ or $m_{ji}=0$.
- If R is antisymmetric relation, then for different vertices i and j there cannot be an edge from vertex i to vertex j and an edge from vertex j to vertex i
- At least one directed relation and one way

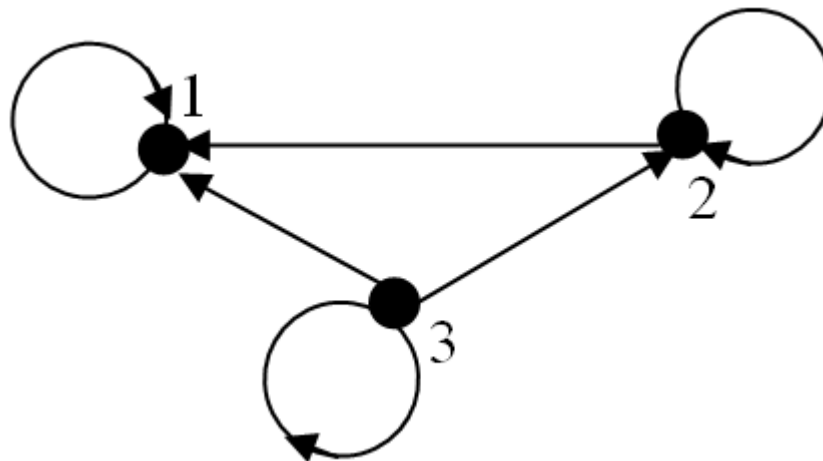


Example

- Let R be a relation on $A = \{1, 2, 3\}$ defined as $(a, b) \in R$ if $a \geq b$, $a, b \in A$ is an antisymmetric relation because for all $a, b \in A$, $(a, b) \in R$ and $a \neq b$, then $(b, a) \notin R$, for example

$$(3, 2) \in R \text{ but } (2, 3) \notin R$$

$$(3, 3) \in R \text{ and } (3, 3) \in R \text{ implies } a = b$$





example

- The relation R on $X = \{ 1, 2, 3, 4 \}$ defined by,

$$(x, y) \in R \quad \text{if } x \leq y, x, y \in X$$

$$\begin{aligned} (1, 2) &\in R \\ (2, 1) &\notin R \end{aligned}$$

	1	2	3	4
1	1	1	1	1
2	0	1	1	1
3	0	0	1	1
4	0	0	0	1

antisymmetric



example

- The relation $R = \{ (a,a), (b,c), (c,b), (d,d) \}$ on $X = \{ a, b, c, d \}$

$(b,c) \in R$
 $(c,b) \in R$

	a	b	c	d
a	1	0	0	0
b	0	0	1	0
c	0	1	0	0
d	0	0	0	1

not antisymmetric



example

- The relation

$$R = \{ (a,a), (b,b), (c,c) \}$$

on $X = \{ a, b, c \}$

- R has no members of the form (x,y) with $x \neq y$, then R is **antisymmetric**

	a	b	c
a	1	0	0
b	0	1	0
c	0	0	1



example

- Antisymmetric
- Reflexive
- Symmetric

	a	b	c
a	1	0	0
b	0	1	0
c	0	0	1

- “Antisymmetric” is not the same as “not symmetric”

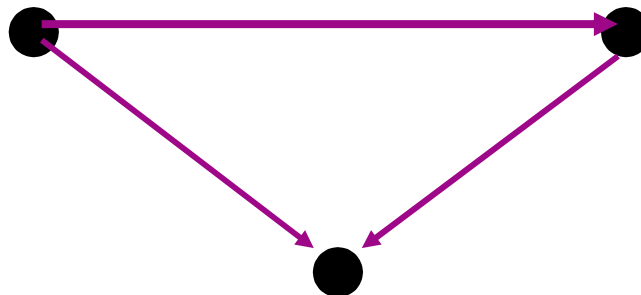


UTM
UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Antisymmetric Relations

- The digraph of an antisymmetric relation has at most one directed edge between each pair of vertices.
- Example





UTM
UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Asymmetric

A relation R on set A is asymmetric if whenever aRb , then $b \not R a$.

$$\forall x, y \in X, (x, y) \in R \rightarrow (y, x) \notin R$$

In this sense, a relation is asymmetric if and only if it is both antisymmetric and irreflexive.



Asymmetric

- The matrix $M_R = [m_{ij}]$ of an asymmetric relation R satisfies the property that
If $m_{ij} = 1$ then $m_{ji} = 0$
 $m_{ii} = 0$ for all i (the main diagonal of matrix M_R consists entirely of 0's or otherwise)
- If R is asymmetric relation, then the digraph of R cannot simultaneously have an edge from vertex i to vertex j and an edge from vertex j to vertex i
- All edges are “one way street”



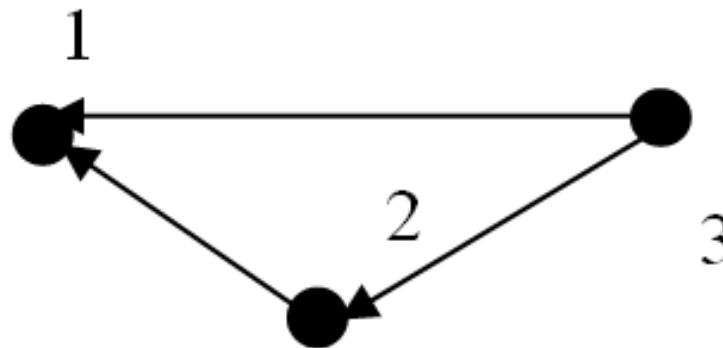
Example

- Let R be the relation on $A = \{1, 2, 3\}$ defined by $(a, b) \in R$ if $a > b$, $a, b \in A$ is an asymmetric relation because,

$(2, 1) \in R$ but $(1, 2) \notin R$

$(3, 1) \in R$ but $(1, 3) \notin R$

$(3, 2) \in R$ but $(2, 3) \notin R$





Not Symmetric

- Let R be a relation on a set A .
- Then R is called ***not symmetric***, if for all $a, b \in A$, if $(a, b) \in R$, there exist $(b, a) \notin R$.

$$\exists a, b \in A, (a, b) \in R \rightarrow (b, a) \notin R$$



Not Symmetric and Not Antisymmetric

- Let R be a relation on a set A . Then R is called ***not symmetric*** and ***not antisymmetric***, if for all $a, b \in A$, if $(a, b) \in R$, there exist $(b, a) \notin R$ and if $(a, b) \in R$, there exist $(b, a) \in R$.

$$\exists a, b \in A, (a, b) \in R \rightarrow (b, a) \in R$$

AND

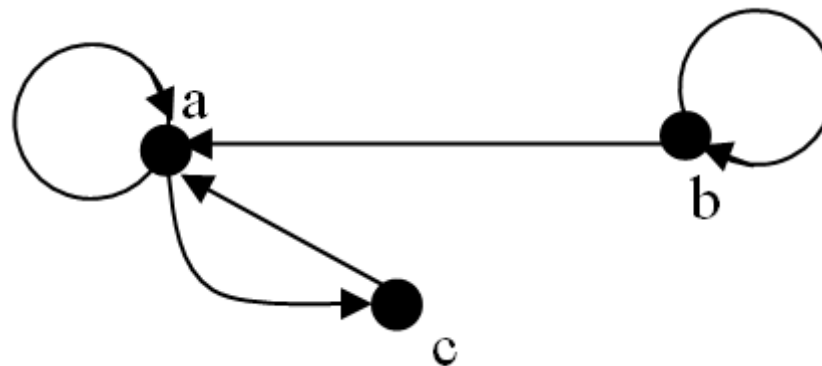
$$\exists a, b \in A, (a, b) \in R \rightarrow (b, a) \notin R$$



Example

- Relation $R = \{(a, c), (b, b), (c, a), (b, a), (a, a)\}$ on $A = \{a, b, c\}$ is not symmetric and not antisymmetric relation because there is,

$(a, c), (c, a) \in R$ and also $(b, a) \in R$ but $(a, b) \notin R$





UTM
UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

1. Let $A = \mathbb{Z}$, the set of integers and let $R = \{(a, b) \in A \times A \mid a < b\}$.
So that R is the relation “less than”.

Is R symmetric, asymmetric or antisymmetric?

2. Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1, 2), (2, 2), (3, 4), (4, 1)\}$

Determine whether R symmetric, asymmetric or antisymmetric.



Example

Question 1

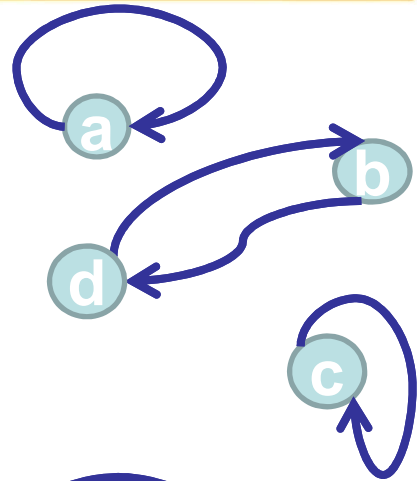
- **Symmetric** : If $a < b$, then it is not true that $b < a$, so R is not symmetric
- **Assymmetric** : If $a < b$ then $b > a$ (b is greater than a), so R is assymmetric
- **Antisymmetric** : If $a \neq b$, then either $a > b$ or $b > a$, so R is antisymmetric

Question 2

- R is not symmetric since $(1,2) \in R$, but $(2,1) \notin R$
- R is not asymmetric, since $(2,2) \in R$
- R is antisymmetric, since $a \neq b$, either $(a,b) \in R$ or $(b,a) \notin R$

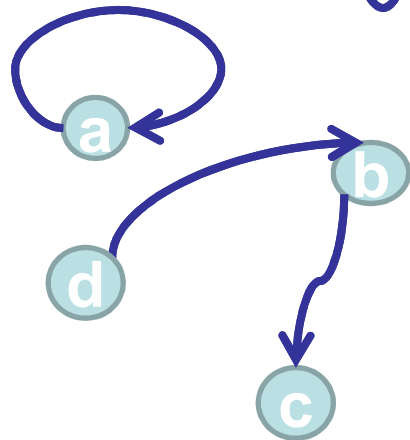
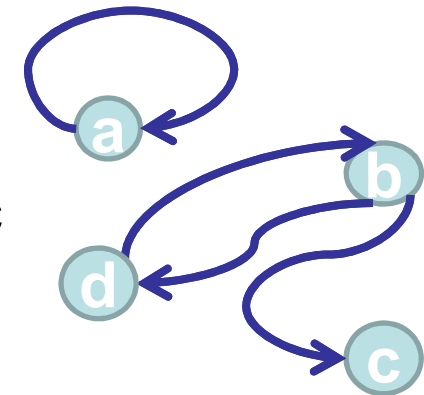


Summary on Symmetric



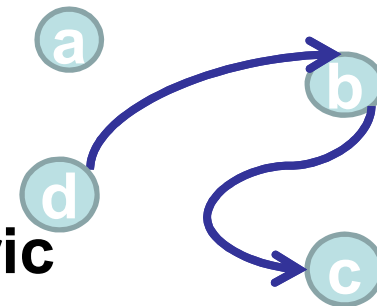
Symmetric

Not Symmetric



Antisymmetric

Asymmetric





Transitive Relations

- A relation R on a set X is called transitive if for all $x, y, z \in X$,
if (x, y) and $(y, z) \in R$ then $(x, z) \in R$
- It is often convenient to say what it means for a relation to be not transitive.
- A relation R on X is **not transitive** if there exists x, y , and z in X so that xRy and yRz , but $\cancel{xRz}$. If such x, y , and z do not exist, then R is transitive.



Example

- The relation $R = \{ (a,a), (b,c), (c,b), (d,d) \}$ on $X = \{ a, b, c, d \}$ is not transitive.
- (b,c) and $(c,b) \in R$, but $(b,b) \notin R$.



Transitive Relations

- Let M_R be the matrix of relation R . Let,
 $M_R \otimes M_R = N.$ ($\otimes$ Boolean product)

$$M_R = [m_{ij}] \text{ and } N = [n_{ij}]$$

The relation R is **transitive** if and only if the following is true:

$$\forall i \forall j, \text{ if } (n_{ij} = 1) \text{ then } (m_{ij} = 1)$$

The relation R is **not transitive** if and only if:

$$\exists i \exists j (n_{ij} = 1) \wedge (m_{ij} = 0)$$



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Boolean Algebra

+	1	0
1	1	1
0	1	0

.	1	0
1	1	0
0	0	0

Example

Let R be a relation on $A=\{1,2,3\}$ is defined by $(a,b) \in R$ if $a \leq b$, $a,b \in A$. Find R . Is R a transitive relation?

Solution:

$$R=\{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$$

R is a transitive relation because

$$(1,2) \text{ and } (2,2) \in R, (1,2) \in R$$

$$(1,2) \text{ and } (2,3) \in R, (1,3) \in R$$

$$(1,3) \text{ and } (3,3) \in R, (1,3) \in R$$

$$(2,2) \text{ and } (2,3) \in R, (2,3) \in R$$

$$(2,3) \text{ and } (3,3) \in R, (2,3) \in R$$



example

The matrix of relation M_R ,

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$\forall i \forall j, \text{ if } (n_{ij} = 1)$
then } (m_{ij} = 1)

The product of boolean,

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Note that, $(1,2) \in R$, $(2,3) \in R$, $(1,3) \in R$



Example

- Consider the following relations on the set $\{1, 2, 3\}$

$$R1 = \{ (1,1), (1,2), (2,3) \}$$

$$R2 = \{ (1,2), (2,3), (1,3) \}$$

- Which of them is transitive?



Example

$$R1 = \{ (1,1), (1,2), (2,3) \}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\forall i \forall j$, if $(n_{ij} = 1)$ then $(m_{ij} = 1)$

$$(n_{13} = 1) \wedge (m_{13} = 0)$$

$(1,2)$ and $(2,3) \in R$, $(1,3) \notin R$

Not transitive



Example

$$R_2 = \{ (1,2), (2,3), (1,3) \}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & 1 & \textcircled{1} \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\forall i \forall j$, if $(n_{ij} = 1)$ then $(m_{ij} = 1)$

$(n_{13} = 1)$ then $(m_{13} = 1)$

$(1,2)$ and $(2,3) \in R$, $(1,3) \in R$

R is a transitive relation



Example

The relation R on $A=\{a,b,c,d\}$ is $R=\{(a,a), (b,b), (c,c), (d,d), (a,c), (c,b)\}$ is not transitive. The matrix of relation M_R ,

$$M_R = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 1 & \textcircled{0} & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

The product of boolean,

$$(n_{12} = 1) \wedge (m_{12} = 0)$$

$$\begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix} \otimes \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix} = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 1 & \textcircled{1} & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Note that, (a,c) and $(c,b) \in R$, $(a,b) \notin R$



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

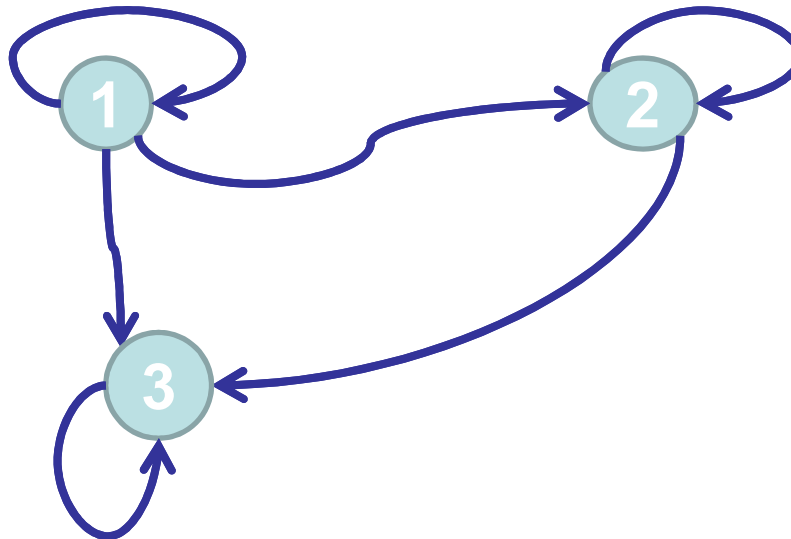
Transitive Relations

In the digraph of R , R is a transitive relation if and only if there is a directed edge from one vertex a to another vertex b , and if there exists a directed edge from vertex b to vertex c , then there must exist a directed edge from a to c



$$R = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$$

The diagraph:





UTM
UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Equivalence Relations

- A relation R that is reflexive, symmetric and transitive on a set X is called an equivalence relation on X .



Example

Let $R=\{(1,1), (1,3), (2,2), (3,1), (3,3)\}$ on $\{1,2,3\}$, the matrix of the relation M_R ,

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

All the main diagonal matrix elements are 1 and the matrix is reflexive

Example

Let $R=\{(1,1), (1,3), (2,2), (3,1), (3,3)\}$ on $\{1,2,3\}$, the matrix of the relation M_R ,

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

All the main diagonal matrix elements are 1 and the matrix is reflexive



example

The transpose matrix M_R , M_R^T is equal to M_R , so R is symmetric

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{matrix} \quad M_R^T = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

The product of Boolean show that the matrix is transitive.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$



Example

- The relation, $R = \{ (1,1), (1,3), (1,5), (2,2), (2,4), (3,1), (3,3), (3,5), (4,2), (4,4), (5,1), (5,3), (5,5) \}$ on $\{ 1,2,3,4,5 \}$

- Reflexive?
- Symmetric?
- Transitive?

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$



■ Reflexive?

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Reflexive ✓



■ Symmetric?

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Symmetric ✓



■ Transitive?

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$



example

- Reflexive
- Symmetric
- Transitive



Equivalence relation

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$



example

- The relation R on $X = \{ 1, 2, 3, 4 \}$, defined by
 $(x, y) \in R$ if $x \leq y, x, y \in X$
- Not symmetric
 - $(2, 3) \in R$ but $(3, 2) \notin R$
- R is not equivalence relation on X .



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Partial Orders

- A relation, R on a set X is called a **partial order** if R is reflexive, antisymmetric, and transitive.

Example

Let R be a relation on a set $A=\{1,2,3\}$ defined by $(a,b)\in R$ if $a\leq b$, $a,b\in R$.

$$R = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$$

R is reflexive, antisymmetric and transitive.

So R is a partial order relation.



Example

- The relation R defined on the positive integers by
 $(x,y) \in R$ if x divides y (evenly)

is reflexive, antisymmetric and transitive
- R is a partial order.